

Tata Steel Safety Manual

Version 2.0 | 2024 | Plant Operations Division

Section 1: Safety Vision & Policy

Tata Steel is committed to "Zero Harm" — a workplace free from injuries, occupational illness, and environmental incidents. Every employee, contractor, and visitor is responsible for safety. Safety takes precedence over production targets, deadlines, and cost considerations at all times.

Core Safety Principles:

- All injuries and occupational illnesses are preventable
- Safety is everyone's responsibility, not just the Safety Department
- Working safely is a condition of employment
- Management is accountable for providing a safe workplace
- Employees have the right and duty to stop unsafe work

Section 2: General Safety Rules

All employees must follow these rules at all times:

- Always wear Personal Protective Equipment (PPE) in plant areas
- Never operate equipment without proper training and authorization
- Report all accidents, incidents and near misses immediately to supervisor
- No smoking in plant premises except designated areas
- Mobile phones prohibited in operational areas
- Always use designated walkways; never cross conveyor belts or rail tracks
- Do not remove or bypass safety guards on machinery
- Horseplay, running, and unsafe behavior strictly prohibited
- Report any unsafe condition immediately, even if not your area

Section 3: Personal Protective Equipment (PPE)

Mandatory PPE by area:

Plant Operations:

- Safety helmet (IS 2925 certified)
- Safety shoes (steel toe cap, IS 15298)
- High visibility jacket
- Safety goggles
- Ear plugs in high noise areas (above 85 dB)

Chemical Handling Areas:

- Chemical resistant gloves (nitrile or neoprene)
- Face shield
- Chemical resistant apron
- Respiratory mask (P100 rated)

Hot Work Areas (Welding, Cutting, Furnace):

- Fire retardant clothing
- Welding helmet with appropriate shade
- Leather gloves
- Leather boots/spats

Confined Space Entry:

- Gas detector (4-gas monitor)
- Full body harness with retrieval line
- Communication device
- Standby person mandatory

PPE Inspection Schedule:

PPE Item	Inspection Frequency	Replacement Trigger	
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Safety helmet	Daily visual check	Cracks, dents, 3 years age	
Safety shoes	Weekly	Sole wear, toe cap exposure	
Harness	Before each use	Frayed straps, damaged buckles	
Gas detector	Before each use (bump test)	Failed calibration	

Section 4: Chemical Spill Procedure

Immediate steps during chemical spill:

1. Alert nearby workers immediately by shouting and activating alarm
2. Press emergency alert button (red button on every pillar)
3. Call emergency helpline: 1800-TATA-SOS
4. Do NOT attempt cleanup alone
5. Move to safe assembly point (upwind of spill)
6. Wait for emergency response team
7. Report to supervisor with details: chemical name, quantity, location

Do not:

- Ignore small spills — they can spread or generate fumes
- Clean up without proper PPE
- Delay reporting under any circumstances
- Allow spilled material to enter drains

Spill Kit Locations:

Spill kits are located at every chemical storage area, loading dock, and laboratory. Each kit contains absorbent material, neutralizing agents, disposal bags, and PPE.

Chemical-Specific Notes:

| Chemical | Hazard | Special Precaution |

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| Sulfuric Acid | Corrosive, toxic fumes | Never add water to acid |

| Caustic Soda | Severe burns | Flush with water 15+ minutes |

| Hydrochloric Acid | Toxic vapors | Use in ventilated area only |

| Coal Tar | Carcinogenic | Avoid skin contact entirely |

Section 5: Fire Emergency Procedure

****Steps during fire:****

1. Raise alarm immediately (pull red handle)
2. Call fire emergency: Extension 999
3. Evacuate using nearest emergency exit
4. Do NOT use elevators
5. Assemble at designated muster point
6. Do NOT re-enter building until clearance
7. Account for all team members at muster point

****Fire extinguisher locations:****

- Every 15 meters in plant areas
- Near all electrical panels
- At all exit points
- Inside vehicle cabins (mobile equipment)

****Fire Extinguisher Types & Usage:****

| Type | Color Code | Use For |

|---|---|---|

| Water (CO2 propelled) | Red | Wood, paper, textile fires |

| Foam | Cream | Flammable liquid fires |

CO2	Black	Electrical equipment fires
Dry Chemical Powder	Blue	Multi-purpose, gas fires
Sand Bucket	-	Small metal fires (do not use water)

****PASS Technique for Extinguisher Use:****

- ****P**ull** the safety pin
- ****A**im** at the base of the fire
- ****S**queeze** the handle
- ****S**weep** side to side

Section 6: Electrical Safety

- Never touch electrical equipment with wet hands
- Report damaged cables immediately
- Only authorized electricians to do electrical work
- Lockout/Tagout (LOTO) procedure mandatory before maintenance
- Minimum safe distance from high voltage: 3 meters (up to 11kV), 6 meters (above 11kV)
- Never use damaged extension cords or multi-plugs
- All portable tools must have earth leakage protection

****LOTO Procedure Steps:****

1. Notify all affected personnel
2. Identify all energy sources (electrical, hydraulic, pneumatic)
3. Shut down equipment using normal procedure
4. Isolate energy sources (open breakers, close valves)
5. Apply lock and personal tag
6. Verify zero energy state (test before touch)
7. Perform maintenance work
8. Remove locks only by person who applied them
9. Verify area clear before re-energizing

Section 7: Working at Heights

- Safety harness mandatory above 1.8 meters
- Inspect harness before every use (straps, buckles, stitching)
- Never work at heights alone
- Secure all tools to prevent falling (tool lanyards mandatory)
- Report damaged safety equipment immediately
- Scaffolding must be inspected and tagged before use
- Weather check mandatory — no height work during lightning or high winds (above 40 km/h)

Fall Protection Hierarchy:

1. Elimination — avoid working at height where possible
2. Prevention — guardrails, covers, barriers
3. Restraint — harness with restraint lanyard (prevents reaching edge)
4. Arrest — harness with shock-absorbing lanyard (limits fall distance)

Section 8: Material Handling & Crane Safety

- Never exceed rated load capacity of any lifting equipment
- Inspect slings, chains, and hooks before each lift
- No personnel under suspended loads at any time
- Crane operators must be certified and current on license
- Use tag lines to control load movement
- Communicate using standard hand signals or radio
- Daily pre-use inspection checklist mandatory for all cranes

Common Material Handling Hazards:

| Hazard | Prevention |

|---|---|

| Manual lifting injuries | Use mechanical aids; team lifts for >25kg |

| Falling objects | Barricade area below overhead work |

| Pinch points | Keep hands clear of load paths |

| Unstable stacking | Follow stacking height limits, use pallets |

Section 9: Confined Space Entry

A confined space is any area with limited entry/exit, not designed for continuous occupancy, with potential for hazardous atmosphere.

****Examples at Tata Steel:****

- Storage tanks and silos
- Pits and trenches
- Ducts and flues
- Vessel interiors

****Entry Requirements:****

1. Valid confined space entry permit
2. Atmosphere testing (oxygen, flammable gas, toxic gas) before entry
3. Continuous monitoring during occupation
4. Ventilation if required
5. Standby attendant outside at all times
6. Rescue plan and equipment ready
7. Communication method established

Section 10: Emergency Contacts

Emergency	Contact
Fire	Extension 999
Medical	Extension 998
Security	Extension 997
General Emergency	1800-TATA-SOS
Safety Officer	Extension 201
Environment Cell	Extension 205
Ambulance	Extension 996

Section 11: Incident Reporting & Investigation

All incidents — including near misses — must be reported within 2 hours of occurrence.

****Reporting Process:****

1. Provide first aid / secure the area
2. Report to immediate supervisor verbally
3. Fill incident report form on Safety Portal within 24 hours
4. Preserve evidence (do not disturb scene unless danger persists)
5. Investigation team assigned within 48 hours for serious incidents
6. Corrective actions tracked to closure

****Severity Classification:****

Category	Examples	Investigation Timeline
Near Miss	Object fell, no injury	3 days
First Aid Case	Minor cut, bruise	3 days
Medical Treatment	Stitches, fracture	7 days
Lost Time Injury	Unable to work next shift	15 days
Fatality / Major	Death, permanent disability	Immediate, senior management led